

Question number	Answer	Additional guidance	Mark
3(a)	An answer that includes the following points: • (speciation) formation of one {new / different} species (from a pre-existing one) (1) • (sympatric) while both continue to inhabit the same location (1)	NB IGNORE explanations unless contradictory DO NOT ACCEPT formation of {two species / subspecies} ACCEPT without a physical barrier	(2)

Question number	Answer	Mark
3(b)	The only correct answer is D <i>A is incorrect because separation of original species does not take place</i> <i>B is incorrect because some of the original species do not move away from each other</i> <i>C is incorrect because some of the original species do not move away from each other</i>	(1)

Question number	Answer	Additional guidance	Mark
3(c)(i)	An answer that includes two of the following points: • ability to detect a different {colour / smell / shape} (of fruit) (1) • are able to {feed on / lay eggs in / penetrate} apples (1) • flies {lay their eggs / mate } at different {times of year / temperatures} (1) • {eggs/ maggots / flies} need to survive in different temperatures (1)	ACCEPT have enzymes to feed on apples ACCEPT higher / lower temps ACCEPT higher / lower temps resistance to pesticides	(2)

Question number	Answer	Additional guidance	Mark
3(c)(ii)	An explanation that includes three of the following points: • mutation (in the DNA / existing gene) (1) • mutation was present in the gametes of {either one / both} flies (1) • (this) {gene / allele} passed onto the offspring (1) • {presence of apples / lack of berries} acted as a selection pressure (1)	DO NOT ACCEPT caused by selection pressure	(3)

Question number	Answer	Additional guidance	Mark
3(c)(iii)	An explanation that includes two of the following points: - so that there was no (interspecific) <u>competition</u> for {food (maggots / adults) / space (to lay eggs)} (1) - maggots (of both species) are more likely to {grow / develop / survive} (if less competition) (1) - increasing the numbers of (both types of) flies (if less competition) (1)	ACCEPT hatched eggs ACCEPT hatched eggs	(2)

Question number	Answer	Additional guidance	Mark
3(c)(iv)	An answer that includes two of the following points: - eggs {laid / hatched} at different times of the year (1) - so the {adults / flies} were present at different times of the year (1) - different breeding behaviours / incompatible {gametes / genitalia} (1)	ACCEPT mating occurred at different times of the year ACCEPT anatomically incompatible do not recognise each other	(2)